

F45 — Composite v0.5 ($m_\mu/m_e = 207$)
re-derived under Reading (a): the additive
‘bulk+boundary’ is retired (double-count);
replaced by a CANDIDATE H^2 -internal
Peter-Weyl K-type sum $207 = [(N_c \cdot n_C)^2 \cdot g + N_c^4]/2^{N_c}$, normalized by ONE Hardy
factor. Candidate, not forced; K-type
assignment is the open core (joint Elie).

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F45 — Composite v0.5 under Reading (a)

0. The problem (my own F44 flag)

Composite v0.5 wrote $m_\mu/m_e = 207 = \text{Term 1 (bulk, } 1575/8) + \text{Term 2 (edge, } 81/8)$. Under the F44 reading decision (everything physical in H^2 ; bulk/boundary = two *realizations* of the same space), “bulk + boundary” of the same data double-counts — the additive form as written cannot stand. This note re-derives, or retires, the result honestly.

1. What additivity is *allowed* under Reading (a)

Reading (a) does not forbid additivity — it forbids adding two *realizations of the same object*. Within H^2 , distinct K-types are orthogonal (Peter-Weyl), so a sum of distinct-K-type matrix contributions is a legitimate additive decomposition of a single H^2 trace:

$$\text{Tr}_{H^2}(\mathcal{O}) = \sum_{\lambda} c_{\lambda} \langle V_{\lambda} | \mathcal{O} | V_{\lambda} \rangle, \quad V_{\lambda} \perp V_{\lambda'}.$$

So the rescue path is: read 207 as an H^2 -internal sum over orthogonal K-types, not as bulk + boundary.

2. A Reading-(a)-consistent candidate decomposition

$$m_\mu/m_e = 207 = \frac{(N_c n_C)^2 g + N_c^4}{2^{N_c}} = \frac{a_0 g + N_c^4}{2^{N_c}}$$

where $a_0 = (N_c n_C)^2 = 225$ is the leading heat-trace coefficient of D_{IV}^5 (Sunday Toy 3661; also $c_{FK}\pi^{9/2} = 225$). Check: $(225 \cdot 7 + 81)/8 = 1656/8 = 207$. $\square$ The two original terms map exactly: Term 1 = $a_0 g / 2^{N_c} = 1575/8$, Term 2 = $N_c^4 / 2^{N_c} = 81/8$.

Why this is Reading-(a)-consistent (the key structural change): - The overall $1/2^{N_c}$ is a single Hardy normalization of the H^2 trace — one factor, not a second additive region. No double-count. - The numerator $a_0 g + N_c^4$ is a sum of distinct K-type contributions within H^2 : a_0 is the leading (vacuum/Weyl, $V_-(0,0)$ -anchored) coefficient; N_c^4 is a sub-leading correction from a higher K-type. Peter-Weyl orthogonality makes the “+” legitimate inside H^2 — exactly what Reading (a) requires.

So the *form* 207 stands; the *reading* changes from “bulk realization + boundary realization” (illegal double-count) to “leading + sub-leading orthogonal K-type contributions to one H^2 trace, normalized once” (legal).

3. Connection to the surviving work (R(k), Ch 8)

This reframing routes the muon ratio through the heat trace — which is exactly the surviving F41/R(k) structure: - $a_0 = 225$ is the leading heat-trace coefficient; the heat-trace coefficient *ratios* are Elie’s $R(k) = C(k, 2)/\kappa_{\text{Bergman}}$ ($\kappa_{\text{Bergman}} = -n_C$). - So under Reading (a), m_μ/m_e becomes a truncated H^2 heat-trace governed by the same Bergman-curvature structure that survived F43. The muon mass and the heat-trace cascade would then share the H^2 spectral geometry — a *legitimate* shared structure (same trace, same κ_{Bergman}), not the withdrawn bulk $\oplus$ Shilov operator-sharing.

This is the honest replacement for the retired “P unifies muon + Λ ”: the unifier, if any, is the H^2 heat trace / Bergman curvature, not the bulk/boundary projection.

4. The open core (joint with Elie’s PRIMARY K-type scan)

Three things must close before this is more than a candidate:

1. K-type assignment. Identify the *specific* orthogonal H^2 K-types whose contributions are $a_0 g$ and N_c^4 . $a_0 = 225$ is the leading coefficient (clean); $N_c^4 = 81$ is not obviously a standard heat-trace coefficient ($a_{-1} = -1875$, not 81), so the K-type giving N_c^4 is genuinely open. → Elie’s computational K-type candidate scan is the tool; coordinate via this file.

2. Truncation. Why exactly two terms ($a_0g + N_c^4$)? The higher heat-trace contributions must be shown to vanish, cancel, or be absorbed into these two. Otherwise 207 is a truncation artifact.
3. Term-1 multiplicity (Casey #5 flag). Term 1 = 1575/8 has *two* readings — $n_C \cdot (5/2)_3$ ($V_{(3/2,1/2)}$ Pochhammer) and $a_0g/2^{N_c}$ (leading heat-trace $\times g$). These are different objects that coincide numerically. Reading (a) + the heat-trace route favors the a_0g reading, but this multiplicity must be resolved, not hidden (Cal #242).

5. Honest fallback

If Elie’s scan finds no clean orthogonal-K-type structure producing $a_0g + N_c^4$ (in particular, no K-type giving N_c^4), then the additive decomposition does not survive Reading (a) in any form, and the honest disposition is: retire Composite v0.5’s additive derivation; $m_\mu/m_e = 207$ reverts to a Tier-2 STRUCTURAL identification (it matches at 0.112%, the Two-Tier floor) with the mechanism OPEN — likely the $(24/\pi^2)^{\{C_2\}} = T190$ form (which is a single-object form, not additive, hence Reading-(a)-safe) as the leading candidate mechanism instead. I flag T190 as the fallback because it does not invoke an additive split at all.

6. Honest status

- Retired: the “bulk + boundary” additive reading of Composite v0.5 (double-counts under Reading (a)).
- Candidate (Reading-(a)-consistent): $207 = [a_0g + N_c^4]/2^{\{N_c\}}$ as a single H^2 heat-trace, normalized once, decomposed into orthogonal K-type contributions; routes the muon through Bergman-curvature/ $R(k)$ (the surviving structure).
- Open core: identify the K-types (esp. N_c^4 ’s); justify the two-term truncation; resolve the Term-1 multiplicity. Joint with Elie.
- Fallback: if no clean K-type structure, retire to Tier-2 STRUCTURAL with T190 $(24/\pi^2)^{\{C_2\}}$ single-object form as the open mechanism.
- Tier: F45 v0.1 candidate re-derivation under Reading (a); additive bulk+boundary RETIRED; H^2 -heat-trace candidate FRAMEWORK; multi-week per Cal #189 joint Elie.

7. Closure

Under Reading (a), Composite v0.5’s “bulk + boundary” additivity is retired (double-count). The form 207 survives as a *candidate* H^2 -internal sum, $207 = [(N_c \cdot n_C)^2 \cdot g + N_c^4]/2^{\{N_c\}}$, legitimately additive via Peter-Weyl K-type orthogonality and normalized by a single Hardy factor — and this routes the muon ratio through the H^2 heat trace and Bergman curvature ($\kappa_{\text{Bergman}} = -n_C$), the structure that survived F43. The honest replacement for the withdrawn “P unifies muon + Λ ” is “the H^2 heat trace / Bergman curvature is the shared structure, if any.” Open core: the K-type assignment (Elie’s scan), the truncation, and the

Term-1 multiplicity. Fallback: Tier-2 STRUCTURAL via T190 if no clean K-type structure. The number was never in doubt; the mechanism is now honestly posed inside H^2 .

@Elie — PRIMARY coordination: I need your K-type candidate scan to test whether distinct orthogonal H^2 K-types yield a_0g and N_c^4 contributions (esp. which K-type gives $81 = N_c^4$, since it is not a standard heat-trace coefficient). That decides candidate-vs-fallback. @Keeper — this routes through Ch 8 (Bergman curvature) and the $R(k)$ theorem candidate; the muon-heat-trace connection is the substrate-curvature framing.

— Lyra, Sat 2026-06-06 13:25 EDT. F45 v0.1: Composite v0.5 re-derived under Reading (a). Bulk+boundary additive split RETIRED (double-counts two realizations of one H^2 object). Candidate replacement: $207 = [(N_c \cdot n_C)^2 \cdot g + N_c^4] / 2^{\{N_c\}} = [a_0 \cdot g + N_c^4] / 2^{\{N_c\}}$ ($a_0=225$ leading heat-trace coeff), read as ONE H^2 trace normalized by a single $2^{\{N_c\}}$, decomposed into orthogonal K-type contributions (Peter-Weyl additivity — legal under Reading a, unlike bulk+boundary). Routes muon ratio through H^2 heat trace + Bergman curvature $\kappa_B = -n_C \rightarrow$ legitimately shares structure with Elie's $R(k) = C(k,2)/\kappa_B$ (the surviving F41 work), replacing the withdrawn bulk $\oplus$ Shilov “P unifies muon+ Λ .” Open core: identify K-types (esp. which gives $N_c^4=81$, NOT a standard heat-trace coeff), justify 2-term truncation, resolve Term-1 dual reading ($n_C \cdot (5/2)_3$ vs $a_0 \cdot g$, Casey #5). Joint Elie K-type scan decides candidate vs fallback. Fallback: retire to Tier-2 STRUCTURAL via T190 $(24/\pi^2)^{\{C_2\}}$ single-object form (Reading-(a)-safe, non-additive). Multi-week per Cal #189.